PROJECT REPORT

Title: NLP-Based News Category Classification Using TF-IDF and Multinomial Naive Bayes

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GitHub Repository:
https://github.com/AnahaShaji/NLP-News-Category-Classification.git

1. Problem Statement

The project aims to automatically classify news articles into different categories using Natural Language Processing (NLP). The system processes the text of a news article and predicts whether it belongs to Tech, Entertainment, Politics, Business, or Sport.

This is a text classification problem because the input is unstructured text and the output is a predefined category. NLP techniques are used to process the text and extract useful features, which are then given to a machine learning model for classification.

2. Objectives

The main objectives of the project are:

1. To preprocess and clean the text of news articles.
2. To convert text into numerical features using TF-IDF.
3. To train a Multinomial Naive Bayes classifier for news category classification.
4. To predict the category of new, unseen news articles.
5. To evaluate the model using accuracy, precision, recall, F1-score, and confusion matrix.
6. To analyze the performance and identify the limitations of the developed system.

3. Dataset

The project uses the BBC News dataset, which contains news articles grouped into five different categories. The dataset is suitable for this project because it provides text documents along with their corresponding category labels.

Item	**Details**
Dataset:	BBC News Dataset
Dataset source:	[BBC Datasets — Original Source]

Total articles:	2,225
Categories:	5
Categories used:	Tech, Entertainment, Politics, Business, Sport
Input:	News article text
Target:	News category

The articles are stored as text files inside separate category folders. During data preparation, the articles are read and combined into a Pandas DataFrame containing the `text` and `category` fields.

The dataset was divided into 80% training data (1,780 articles) and 20% testing data (445 articles). The text was cleaned before being converted into numerical features using TF-IDF.

Dataset Limitations

The dataset has a limited number of categories and contains news articles from a specific collection. Therefore, the model may not perform equally well on news from completely different sources, writing styles, or topics outside these five categories.

4. Methodology

The project follows a simple NLP-based text classification process. The news articles are first cleaned and preprocessed, then converted into numerical features using TF-IDF. These features are given to a Multinomial Naive Bayes model to predict the category of the news article.

Project Workflow

News Articles → Text Preprocessing → TF-IDF → Multinomial Naive Bayes → Prediction → Evaluation

Steps

1. Text Preprocessing: The news text is converted to lowercase, punctuation and numbers are removed, and common stopwords are removed.
2. Train-Test Split: The dataset is divided into 80% training data and 20% testing data.
3. TF-IDF: The cleaned text is converted into numerical TF-IDF features.

4. Model Training: A Multinomial Naive Bayes classifier is trained using the training data.
5. Prediction: The trained model predicts the category of test articles and new articles.
6. Evaluation: The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

This follows the methodology structure suggested in the assignment: Text Data → Preprocessing → TF-IDF → Classification Model → Prediction → Evaluation.

5. Implementation

The project is implemented using Python and common NLP and machine learning libraries. The implementation includes text preprocessing, feature extraction, model training, prediction, and evaluation.

Tools and Libraries Used

- Python – Used to develop the complete NLP application.
- Pandas – Used to store and manage the news articles in a DataFrame.
- OS – Used to access the dataset folders and text files.
- NLTK – Used for English stopword removal.
- Regular Expressions (`re`) – Used to clean punctuation and numbers from the text.
- Scikit-learn – Used for train-test splitting, TF-IDF feature extraction, Multinomial Naive Bayes, and model evaluation.
- Matplotlib and Seaborn – Used to display the confusion matrix.

NLP Techniques Used

The project uses text preprocessing and TF-IDF (Term Frequency-Inverse Document Frequency) to prepare the news articles for classification.

Machine Learning Model

Multinomial Naive Bayes is used as the classification algorithm. It is trained using the TF-IDF features and predicts one of the five news categories.

6. Results

The trained Multinomial Naive Bayes model was evaluated using the 445 testing articles. The model achieved an overall accuracy of 98.43%, showing that it was able to correctly classify most of the news articles into their respective categories.

Model Performance

Metric	Result
Accuracy	98.43%
Test Articles	445
Correct Predictions	438
Incorrect Predictions	7

Category-wise performance

Category	precision	Recall	F1-score
Business	0.99	0.99	0.99
Entertainment	1.00	0.97	0.99
Politics	0.97	1.00	0.98
Sport	0.97	1.00	0.99
Tech	1.00	0.95	0.97

The confusion matrix shows that most of the test articles were correctly classified. Out of 445 articles, 438 were correctly classified and only 7 were misclassified. The Sport category was classified perfectly, with all 102 Sport articles correctly predicted.

The model was also tested with a new news article to demonstrate its ability to classify previously unseen text.

Screenshots

1.Classification Report

15. Evaluate the Classification Model

The classification report is used to evaluate the performance of the model on the test data. It shows the precision, recall, and F1-score for each news category, along with the overall accuracy of the model.

[62] 0s

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

```
               precision    recall  f1-score   support

     business       0.99      0.99      0.99       102
entertainment       1.00      0.97      0.99        77
     politics       0.97      1.00      0.98        84
        sport       0.97      1.00      0.99       102
         tech       1.00      0.95      0.97        80

     accuracy                           0.98       445
    macro avg       0.99      0.98      0.98       445
 weighted avg       0.98      0.98      0.98       445
```

2.Accuracy

16. Calculate Model Accuracy

Accuracy is calculated to measure the overall performance of the classification model. It represents the percentage of news articles that were correctly classified by the model out of the total test articles.

[63] 0s

```
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
print("Accuracy percentage:", round(accuracy * 100, 2), "%")
```

```
Accuracy: 0.9842696629213483
Accuracy percentage: 98.43 %
```

3.New Article Prediction

19.Test with a new article

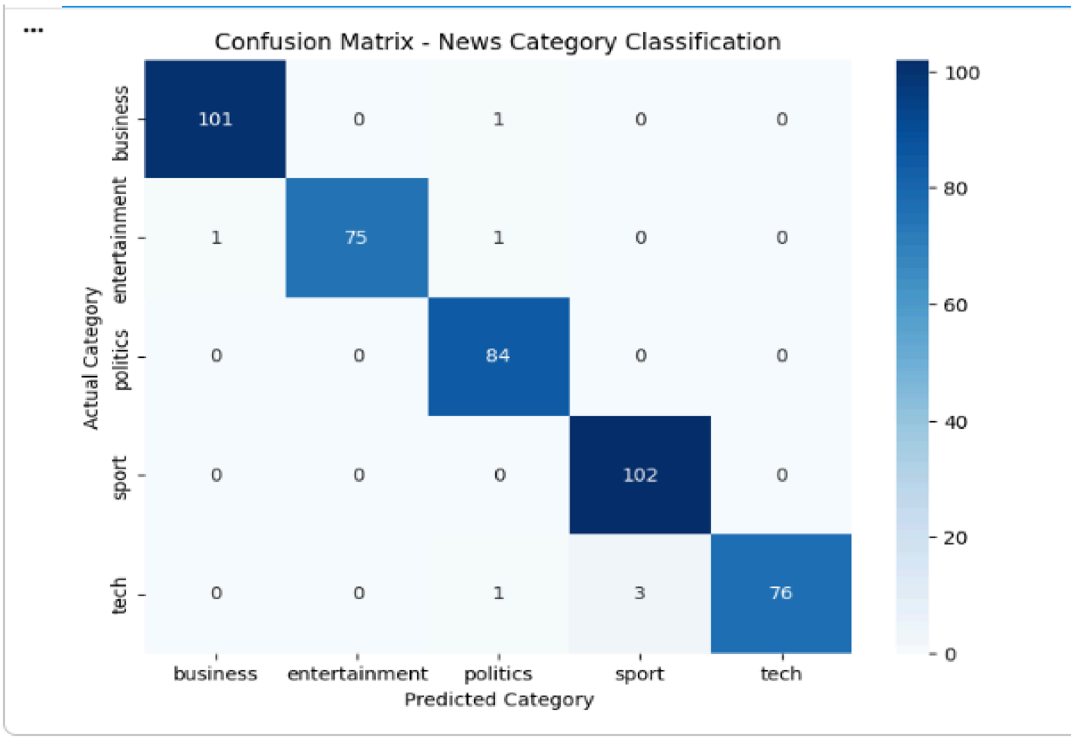
[66]

```
news = """
The football team secured a dramatic victory in the final match.
The players scored two goals in the second half and the coach praised
the team's excellent performance.
"""

result = predict_category(news)
print("Predicted Category:", result)
```

```
Predicted Category: sport
```

4.Confusion Matrix



7. Limitations

Although the model achieved a high accuracy of 98.43%, the project has some limitations:

1. The dataset contains only five news categories, so the model cannot classify articles outside these categories.

2. The model is trained on the BBC News dataset, so its performance may vary on news articles from different sources or writing styles.
3. The system uses basic text preprocessing and TF-IDF, so it may not fully understand the deeper meaning or context of an article.
4. The dataset is relatively small compared with large real-world news datasets.
5. Some articles from different categories may contain similar words, which can occasionally lead to incorrect classification.

8. Future Scope

The project can be improved further in the following ways:

1. More Categories**:** Additional news categories can be added to make the classification system more useful.
2. Larger Dataset**:** A larger and more diverse dataset can be used to improve the model's ability to handle different types of news articles.
3. Advanced NLP Techniques**:** Techniques such as word embeddings and deep learning models can be explored to better understand the meaning and context of news articles.
4. Real-Time Classification**:** The system can be extended to classify news articles collected from online news sources.
5. User Interface**:** A simple web interface can be developed where users can enter a news article and immediately view its predicted category.
6. Model Comparison**:** Other machine learning algorithms can be tested and compared with Multinomial Naive Bayes to identify the most suitable model.

9. Conclusion

The project successfully developed an NLP-based system for automatically classifying news articles into five categories: Business, Entertainment, Politics, Sport, and Tech.

The news articles were cleaned using basic text preprocessing techniques and converted into numerical features using TF-IDF. A Multinomial Naive Bayes classifier was then trained to perform the classification.

The model achieved an overall accuracy of 98.43% on 445 test articles, demonstrating that the approach was effective for the selected dataset. The classification report and confusion matrix also showed strong performance across the five categories.

The project demonstrates how Natural Language Processing and Machine Learning can be combined to automatically organize and classify textual news data.